

Concentration

Concentration (C) is the ratio of the quantity of solute to the quantity of solvent in a solution.

There are many ways of expressing concentration.

Percent Concentrations

Common form, seen on household products, foods, etc.

Percent volume by volume (%V/V)

$$C = \frac{V_{\text{solute}}}{V_{\text{solvent}}} \times 100\% \quad \text{Ex: Vinegar is 5\% acetic acid by volume}$$

Percent mass by volume (%m/V)

$$C = \frac{m_{\text{solute}}}{V_{\text{solvent}}} \times 100\% \quad \text{Ex: There is 3.9g of kool-aid per 2000 mL, or ~0.20\%}$$

Percent mass by mass (%m/m)

$$C = \frac{m_{\text{solute}}}{m_{\text{solvent}}} \times 100\% \quad \text{Ex: A 140g gold necklace may contain 3.0g Au, or ~2.1\%}$$

Someone always asks....



Parts per...

Especially useful for extremely dilute solutions. Used a lot in biology & environmental sciences.

parts per million (ppm) - 1 mg/L - 1 drop in a bathtub

parts per billion (ppb) - 1 drop in a swimming pool

parts per trillion (ppt) - 1 drop in a small lake

Molar Concentration & Molarity

Mainly used in chemistry

The molar concentration is the number of moles of solute per litre of solvent.

$$C = \frac{n}{V}$$

where;

n is the number of moles (mol)

V is the volume in liters (L)

C is the concentration (or molarity) in mol/L or M

5.0 mol of NaCl in 2.0 L of H_2O

$$\begin{aligned} \frac{5.0}{2.0} &= 2.5 \text{ mol/L} \\ &= 2.5 \text{ M} \\ &= [2.5] \end{aligned}$$

Ex) What is the concentration of a saline solution where 3.5g of NaCl is dissolved in 1.50L of water?

$$\begin{aligned} n_{NaCl} &= \frac{3.5 \text{ g}}{58.44 \text{ g/mol}} \\ &= 0.060 \text{ mol} \end{aligned}$$

$$\begin{aligned} C &= \frac{n}{V} \\ &= \frac{0.060}{1.5} \\ &= 0.040 \text{ M} \end{aligned}$$

Ex) How many moles of sugar is required to make a 0.50 M, 250 mL sugar-water solution?

0.25 L

$$C = \frac{n}{V}$$

$$\begin{aligned} n &= CV \\ &= (0.50)(0.25) \\ &= 0.12 \text{ mol} \end{aligned}$$

Homeopathy is a system of medicine which involves treating the individual with highly diluted substances, given mainly in tablet form, with the aim of triggering the body's natural system of healing.

-- The Society of Homeopaths (www.homeopathy-soh.org)

The more dilute the solution, the more effective the cure. (???)

A 1X dilution is one part homeopathic solution in 10 parts water.

A 2X is one part of a 1X solution in 10 parts water. i.e. 1 part in 100

This means that a nX dilution is equivalent to 1 part in 10^n .

So a 9X dilution is 1 part in 10^9 , or 1 part in 1,000,000,000

Homeopathy's first choice for swelling, bruising, and muscular soreness caused by overexertion, **Arnica Montana** is a must for athletes and anyone with high levels of physical activity. Hyland's® handy tablets feature a powerful 30X potency.

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10^{30} parts water

$$\frac{10^{30} \text{ H}_2\text{O}}{6.02 \times 10^{23}} = 1.66 \times 10^6 \text{ mol H}_2\text{O}$$

$$\begin{aligned} m &= nM \\ &= (1.66 \times 10^6 \text{ mol})(18.02 \text{ g/mol}) \\ &= 2.99 \times 10^7 \text{ g} \\ &= 3 \times 10^7 \text{ mL} \\ &= 30,000,000 \text{ mL} \\ &= 30,000 \text{ L} \end{aligned}$$

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